Summary-D1

Core Definitions

- **Proposition (Statement):** A declarative sentence that is either **True** or **False**, but not both.
- **Atomic (Simple) Statement:** A basic statement that does not contain any other statement as a component.
- **Compound Statement:** A statement that contains at least one atomic statement as a component.

Logical Notation

- **Symbols for Propositions:** Typically represented by lowercase letters like p, q, r and s, ...
- **Truth Values:** Denoted as **T** (True) and **F** (False).

Important Logical Operators

Logical operators are used to connect propositions and can be defined by **truth tables**.

Operator	Handle (Natural Language)	Notation (Symbol)	
Negation	*not*	$\neg$	$\backsim$
Conjunction	*and*	$\wedge$	
Disjunction	*or*	$\vee$	
Exclusive-or	*xor (either or)*	$\oplus$	
* Implication	*implies*	" $\Rightarrow$ "	
Biconditional	*if and only if (iff)*	$\Leftrightarrow$	

The Truth Table of Negation

p	$\neg p$
T (True)	**F** (False)
F (False)	**T** (True)

Truth Table for Conjunction

p	q	$p \wedge q$
T	T	**T**
F	T	**F**
T	F	**F**
F	F	**F**

Disjunction (Inclusive OR: $p \vee q$)

p	q	$p \vee q$
T	T	**T**
F	T	**T**
T	F	**T**
F	F	**F**

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2. Exclusive OR (XOR: $p \oplus q$)

The exclusive OR is true only if **exactly one** of the propositions is true. If both are true, or bc are false, the result is false. Think of it like a menu option: "You can have soup or salad" (but r both).

Truth Table:

p	q	$p \oplus q$
T	T	**F**
F	T	**T**
T	F	**T**
F	F	**F**

Note

$p \oplus q$ means "either p or q, but **not both**."

- It is **True** (T) only if the truth values of the two statements are different.
- It is **False** (F) if both statements are the same (both true or both false).

Key Difference

- **Inclusive OR ($p \vee q$):** True when both are true.
- **Exclusive OR ($p \oplus q$):** False when both are true.

Example:

Establish the truth values for the given statements:

- $p: 3 > 1$ True (T)
- $q: 1 = 0$ False (F)
- $r: 2 \times 3 = 6$ True (T)

1. Evaluating $p \oplus q$:

Since the values are different, the result is **True** (T).

2. Evaluating $p \oplus r$:

Since both statements are true, the "not both" rule applies, making the result **False** (F).

The Implication operator ($\Rightarrow$)

It is also known as a **conditional statement**. This operator is used to combine two statements, p and q, into a single logical structure: "If p, then q."

Key Concepts

- **Structure**: Written symbolically as $p \Rightarrow q$.
- **Components**:
  - p: The **condition** or antecedent.
  - q The **consequent** or result.
- **Meaning**: The statement $p \Rightarrow q$ asserts that if p is true, then q **must** also be true. In other words, the truth of p "forces" q to be true.

Example:

Here is a mathematical example to illustrate this logic:

Consider the following statement

r: If the integer a is a multiple of 6, then a is divisible by 2.

Let p: The integer a is a multiple of 6,

q: The integer a is divisible by 2.

Logic: Since any multiple of 6 is even, it is necessarily divisible by 2. Therefore, the implication $p \Rightarrow q$ is true.

Examples:

1. If a number is divisible by 4, then it is even.

2. If a figure is a triangle, then the sum of its interior angles is 180°.

3. If a number ends in 0, then it is divisible by 10.

4. If two lines are parallel, then they never meet.

5. If $x > 5$, then $x > 2$.

6. If a number is prime and greater than 2, then it is odd.

7. If a shape is a square, then it is a rectangle.

Truth Table for $p \Rightarrow q$

In logic, the conditional statement $p \Rightarrow q$ is only **false** in one specific scenario: when the condition p is true, but the result q is false. In all other cases, the statement is considered true.

p	q	$p \Rightarrow q$	Notes
T	T	**T**	The condition is met, and the result happened.
T	F	**F**	**The "Broken Promise":** The condition was met, but the result failed.
F	T	**T**	The condition wasn't met, so the result being true doesn't disprove the rule.
F	F	**T**	Since the condition wasn't met, the rule wasn't "tested," so it remains true by default.

Understanding "Vacuous Truth"

The last two rows (where p is false) are often called **vacuously true**.

Think of it like a promise: *"If it rains (p) then I will bring an umbrella (q)."*

- If it **doesn't rain** (p is False), it doesn't matter if you have an umbrella or not; you haven't broken your promise. Therefore, the statement "If it rains, then I will bring an umbrella" is still technically true.

Note:

The **logical implication**, written as $p \Rightarrow q$. It describes the relationship between a **hypothesis** (p) and a **conclusion** (q).

Common Ways to Say *p* implies *q*

In mathematics and logic, there are many ways to express this relationship in English:

- "If p, then q"
- "p implies q"
- "q if p"
- "p is **sufficient** for q"
- "q is **necessary** for p"
- "p only if q"
- "q whenever p"

Key Takeaways

- **The "False" Hypothesis:** If p is False, the entire implication p implies q is automatically True, regardless of whether q is True or False. This is often called a "vacuously true" statement.
- **Direction Matters:** p implies q is not the same as q implies p. For example, "If it is raining, the ground is wet" is True, but "If the ground is wet, it is raining" might be False (someone could have used a sprinkler).

Problem 01:

hypothesis conclusion

Identify the antecedent and the consequent for the given implications.

(a) *If the triangle has two congruent sides, it is isosceles.*

(b) *The polynomial has at most two roots if it is a quadratic.*

(c) *The data is widely spread only if the standard deviation is large.* $\underbrace{\text{The data is widely spread}}_{p}$ only if $\underbrace{\text{the standard deviation is large}}_{q}$

(d) *The function being constant implies that its derivative is zero.*

(e) *The system of equations is consistent is necessary for it to have a solution.*

(f) *A function is even is sufficient for its square to be even.*

Problem 02: If $\underbrace{\text{a quadrilateral is a square}}_{p}$, then $\underbrace{\text{it is a rectangle}}_{q}$.

$p \Rightarrow q$

q whenever p

q is necessary for p

p is sufficient for q

Write this conditional statement in English using

1. the word "whenever"
2. the phrase "only if"
3. the phrase "is necessary for"
4. the phrase "is sufficient for"

p only if q

Problem 03:

Working with Conditional Statements. Complete the following table:

English Form	Hypothesis	Conclusion	Symbolic Form
If P, then Q.	P	Q	$P \Rightarrow Q$
Q only if P.	Q	P	$Q \Rightarrow P$
P is necessary for Q.	Q	P	$Q \Rightarrow P$
P is sufficient for Q.	P	Q	$P \Rightarrow Q$
Q is necessary for P.	P	Q	$P \Rightarrow Q$
P implies Q.	P	Q	$P \Rightarrow Q$
P only if Q.	P	Q	$P \Rightarrow Q$
P if Q.	Q	P	$Q \Rightarrow P$
If Q then P.	Q	P	$Q \Rightarrow P$
If $\neg Q$, then $\neg P$.	$\sim Q$	$\sim P$	$\sim Q \Rightarrow \sim P$
If P, then $Q \wedge R$.	P	$Q \wedge R$	$P \Rightarrow (Q \wedge R)$
If $P \vee Q$, then R.	$P \vee Q$	R	$(P \vee Q) \Rightarrow R$

Determine the truth value of the following statements

1) $1 > 0$ and $i^2 = -4$ (F)

2) if $\underbrace{1 < 0}_{F}$, then $\underbrace{2^2 + 3^2 = 4^2}_{F}$ (T)

3) if $\underbrace{(\text{either } \underbrace{1 > 0}_{T} \text{ or } \underbrace{3 < 7}_{T})}_{T}$, then $|10| = -7$ (T)

4) $\underbrace{1+2+3+ \cdots 10 = 55}_{T} \Rightarrow \underbrace{4^2 + 7 = 36}_{F}$ (F)

5) if n^2 is even, then n is even (T)

1) Prove that if n^2 is odd, then n is odd (Ex)

2) Prove that $\sqrt{2}$ is a irrational number

Proof of 2):

Suppose $\sqrt{2}$ is a rational number.

Then, $\sqrt{2} = p/q$, where $p, q \in \mathbb{Z}$ and p and q has common factors $(p,q)=1$

$\therefore$ $2 = p^2/q^2$

$\therefore$ $2q^2 = p^2$

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End of Pf2